

DHRSX-Congenital Disorder of Glycosylation (DHRSX-CDG / CDG type 1DD): Comprehensive Disease Report

Disease: DHRSX-Congenital Disorder of Glycosylation **OMIM phenotype:** #301133 (Congenital disorder of glycosylation, type Idd; CDG1DD) **Gene:** *DHRSX* (OMIM 301034) **Category:** *Mendelian (pseudoautosomal-recessive)* **MONDO:*** No disorder-specific term yet; currently subsumed under MONDO:0015286 (congenital disorder of glycosylation)

Summary

DHRSX-CDG is an ultra-rare, severe multisystem congenital disorder of glycosylation (type I) first described in 2024, and it is notable as the first human disease shown to be inherited in a pseudoautosomal-recessive manner. The causal gene, *DHRSX*, resides in pseudoautosomal region 1 (PAR1) at the tips of both the X (Xp22.33) and Y (Yp11.32) chromosomes, so both sexes carry two functional copies that recombine like an autosomal locus. Disease requires biallelic loss-of-function (missense) variants, and because PAR genes escape X-inactivation, males and females are affected equally ([PMID: 38821050](#)).

Mechanistically, the discovery of DHRSX-CDG forced a revision of the textbook dolichol-biosynthesis pathway. The final conversion of polyprenol to dolichol — the lipid carrier essential for N-glycosylation — is not a single reduction performed by SRD5A3 (polyprenol reductase), as long believed, but a **three-step "detour"** involving additional metabolic intermediates. DHRSX catalyzes the first and third steps (an NAD⁺-dependent oxidation of polyprenol to polyprenal, and an NADPH-dependent reduction of dolichal to dolichol), while SRD5A3 is reassigned to catalyze only the intervening second step (reduction of polyprenal to dolichal). Loss of DHRSX function therefore blocks dolichol production, depletes the dolichol-phosphate carrier needed to assemble the lipid-linked oligosaccharide (LLO), and produces protein N-hypoglycosylation with a CDG type I biochemical signature ([PMID: 38821050](#); [PMID: 39395802](#)).

Clinically, the four reported patients (from three unrelated families) present with a severe congenital neurodevelopmental disorder: facial dysmorphism, hypotonia, profound intellectual disability, epilepsy, sensorineural hearing loss, scoliosis, joint contractures, and severe failure to thrive requiring tube feeding. Diagnosis rests on molecular sequencing with attention to the PAR1 locus, supported by a transferrin CDG-I pattern that can normalize with age (and may be normal in some affected individuals) and by polyisoprenoid profiling (elevated polyprenol without polyprenal accumulation, distinguishing DHRSX-CDG from SRD5A3-CDG). No disease-specific therapy exists; management is entirely supportive and multidisciplinary.

Section-by-Section Report

1. Disease Information

Overview. DHRSX-CDG is a congenital disorder of glycosylation of the N-linked type I (assembly/ER) class, caused by biallelic missense variants in *DHRSX*. It manifests as a severe, congenital-onset multisystem neurodevelopmental disorder. It was first delineated by Wilson et al. in 2024, who reported it under the description "a pseudoautosomal-recessive disease presenting as a congenital disorder of glycosylation in patients with missense variants in DHRSX (DHRSX-CDG)" ([PMID: 38821050](#)).

Key identifiers.

Resource	Identifier
OMIM phenotype	#301133 (CDG type Idd; CDG1DD)
OMIM gene (DHRSX)	*301034
OMIM gene (DHRSY paralog)	*400049
HGNC	18399
NCBI Gene	207063
Ensembl	ENSG00000169084
UniProtKB	Q8N5I4 (DHRSX_HUMAN)
MONDO	No disorder-specific term yet; maps to MONDO:0015286 (grouping)
Orphanet	Not yet assigned (2024 first description)
ICD-11	Not yet assigned

Synonyms / alternative names: DHRSX-CDG; CDG-Idd; CDG1DD; congenital disorder of glycosylation type Idd; dolichol synthesis defect (DHRSX). Gene aliases: DHRSY, DHRS5X/DHRS5Y, DHRSXY, SDR46C1, SDR7C6.

Source of information. All clinical knowledge derives from a single aggregated individual-patient report (Wilson et al. 2024) of 4 patients from 3 families, complemented by cell-model and biochemical studies. This is disease-level, primary-literature-derived information rather than EHR/registry aggregation. Importantly, no expanded cohort exists — reported fractions such as 9/10 or 6/7 in the CDG literature belong to a separate PMM2-CDG cohort and must not be attributed to DHRSX-CDG.

2. Etiology

Causal factor: Purely genetic. Biallelic (homozygous or compound-heterozygous) missense variants in *DHRSX* cause loss of enzyme function and blockade of dolichol synthesis. There is no known environmental, infectious, or acquired cause.

Genetic risk factors. The only established risk factor is inheritance of two pathogenic *DHRSX* alleles. Reported pathogenic variants:

Patient	Sex	Genotype
Patient 1	F	Homozygous c.541G>T, p.(Val181Phe)
Patient 2	F	Homozygous c.146C>T, p.(Thr49Met)
Patients 3 & 4 (brothers)	M	Compound-het: c.541G>T p.(Val181Phe) (maternal, X) + c.643C>T p. (Leu215Phe) (paternal, Y)

The brothers' genotype is the direct demonstration of pseudoautosomal-recessive inheritance: one pathogenic allele was transmitted on the maternal X and the other on the paternal Y ([PMID: 38821050](#)).

Environmental / lifestyle / infectious risk factors: None identified or applicable. Consanguinity/homozygosity contributed in the two singleton families (homozygous variants).

Protective factors: None described. (Notably, in the analogous SRD5A3 pathway, an alternative/residual dolichol route can allow partial glycosylation — see Mechanism — but no protective allele or exposure is defined for DHRSX-CDG.)

Gene–environment interactions: None described.

3. Phenotypes

The clinical phenotype (n=4) is a severe congenital multisystem neurodevelopmental disorder. Because the cohort is tiny, frequencies are qualitative.

Phenotype	Type	HPO suggestion	Onset / severity
Facial dysmorphism	Physical manifestation	HP:0001999	Congenital
Muscular hypotonia	Clinical sign	HP:0001252	Congenital, severe
Profound intellectual disability / developmental delay	Neurodevelopmental	HP:0002187 / HP:0001263	Congenital, severe
Epilepsy / seizures	Clinical sign	HP:0001250	Early-onset
Sensorineural hearing loss	Laboratory/clinical	HP:0000407	Congenital/early
Scoliosis	Physical manifestation	HP:0002650	Childhood
Joint contractures	Physical manifestation	HP:0002803	Childhood, progressive
Severe failure to thrive (tube feeding)	Clinical sign	HP:0001508	Neonatal/infantile, severe
Respiratory insufficiency (variable)	Clinical sign	HP:0002093	Variable
Ocular involvement	Physical manifestation	HP:0000478	Congenital/early
Cerebellar atrophy	Imaging/lab abnormality	HP:0001272	Early

Age of onset: congenital / neonatal. **Severity:** severe. **Progression:** static-encephalopathy with progressive orthopedic complications (scoliosis, contractures). **Quality of life:** profound — patients are non-verbal/severely disabled, require tube feeding and multidisciplinary care; daily functioning is severely impaired. Per-phenotype QoL instrument data are not available given the recency and size of the cohort.

4. Genetic / Molecular Information

Causal gene: *DHRSX* (dehydrogenase/reductase X-linked; HGNC:18399; OMIM *301034). A short-chain dehydrogenase/reductase (SDR) family oxidoreductase.

- **Location:** PAR1, Xp22.33, with a functional homolog on Yp11.32; escapes X-inactivation (expressed from both X and Y).
- **Structure:** 7 exons; 330-aa protein (~36.4 kDa) with a Rossmann-fold NAD(P)(H) coenzyme-binding site and a substrate-binding subdomain.
- **Enzyme activities:** polyprenol dehydrogenase and dolichal reductase [NAD(P)+]; dual substrate and cofactor specificity ("DHRSX has a unique dual substrate and cofactor specificity" — [PMID: 38821050](#)).
- **Expression:** widely expressed, highest in pancreas.

Pathogenic variants. All reported disease alleles are **missense** (c.541G>T/p.Val181Phe; c.146C>T/p.Thr49Met; c.643C>T/p.Leu215Phe), classified as pathogenic/likely-pathogenic on functional and segregation grounds. **Functional consequence:** loss of function (loss of dolichol-synthesis enzyme activity). **Somatic vs germline:** germline. **Allele frequencies:** these are private/ultra-rare variants; not established at population scale in gnomAD for this disease.

Modifier genes / epigenetics / chromosomal abnormalities: None established. Of note, an unrelated observation ("DHRSX duplication" fusion) appears as a recurrent structural event in high-risk pediatric B-ALL RNA-seq data ([PMID: 38811988](#)); this is a somatic oncologic finding and is **not** related to the germline DHRSX-CDG phenotype.

5. Environmental Information

Not applicable. DHRSX-CDG is a monogenic inborn error of metabolism with no environmental, lifestyle, toxic, or infectious contributing factors.

6. Mechanism / Pathophysiology

Ordered causal chain (initiating lesion → clinical manifestation)

1. **Biallelic missense variants in *DHRSX*** (on X and/or Y PAR1) **lead to** loss of DHRSX oxidoreductase function.

2. Loss of DHRSX **results in** failure of step 1 of the revised dolichol pathway: polyprenol is no longer oxidized to polyprenal (NAD⁺-dependent dehydrogenase step).
3. Loss of DHRSX **also results in** failure of step 3: dolichal is no longer reduced to dolichol (NADPH-dependent reductase step). (*SRD5A3, the intervening step-2 polyprenal reductase, remains intact but cannot compensate because its substrate/product flow is interrupted upstream and downstream.*)
4. Blockade of steps 1 and 3 **leads to** accumulation of polyprenol (and depletion of dolichol), **without** accumulation of polyprenal — the biochemical fingerprint that distinguishes DHRSX deficiency from SRD5A3 deficiency (which accumulates polyprenal).
5. Dolichol depletion **results in** shortage of dolichol-phosphate, the obligate lipid carrier for activated sugars and the growing oligosaccharide. Accumulated polyprenol-phosphate is a poor substitute substrate for glycosyltransferases.
6. Carrier shortage **leads to** impaired assembly of the lipid-linked oligosaccharide (LLO) on the ER membrane.
7. Defective LLO assembly **results in** protein N-hypoglycosylation (CDG type I pattern), detectable as a hypoglycosylated serum transferrin profile.
8. Systemic hypoglycosylation of many glycoproteins **leads to** the multisystem clinical phenotype — with the developing nervous system, cochlea, skeleton, and other organs particularly vulnerable — manifesting as the congenital neurodevelopmental disorder. (*The tissue-specific vulnerability step is inferred by analogy to other dolichol-pathway CDGs rather than directly demonstrated for DHRSX.*)

The revised three-step "detour" pathway

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Polyprenol
|  STEP 1: DHRSX  (NAD+-dependent oxidation of terminal -OH to aldehyde)
v
Polyprenal
|  STEP 2: SRD5A3 (reduction of C2-C3 alkene) – "polyprenal reductase"
v
Dolichal
|  STEP 3: DHRSX  (NADPH-dependent reduction of aldehyde to alcohol)
v
Dolichol  --->  Dolichol-P  --->  LLO assembly  --->  N-glycosylation

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Chemical rationale. Oxidizing the terminal alcohol of polyprenol to an aldehyde (step 1) activates the adjacent C2–C3 double bond, making it far easier for SRD5A3 to reduce (step 2) than the alkene in polyprenol itself — explaining why a seemingly redundant oxidation/re-reduction "detour" exists. This resolves the long-standing puzzle of how dolichol is made and why SRD5A3-null patients retain residual glycosylation ([PMID: 22304929](#)).

Molecular pathway / GO annotations: - Dolichol biosynthetic process (GO:0019408); dolichol metabolic process (GO:0019348). - Protein N-linked glycosylation via LLO assembly in the ER. - Site of action: endoplasmic reticulum membrane (GO:0005789). - Secondary/independent DHRSX function: positive regulation of autophagy (GO:0010508) — see below.

Metabolic changes / biochemical abnormalities. Elevated polyprenol and polyprenol derivatives, decreased dolichol and derivatives, unchanged polyprenal — the diagnostic metabolite signature of DHRSX deficiency: "Both cell lines showed increased levels of polyprenol and its derivatives, concomitant with decreased levels of dolichol and derivatives, but no change in polyprenal levels, suggesting DHRSX deficiency" ([PMID: 39395802](#)).

A second, independent DHRSX function. Before its dolichol role was known, DHRSX was characterized as "a novel non-classical secretory protein involved in the positive regulation of starvation induced autophagy" ([PMID: 25076851](#)). Overexpression or recombinant GST-DHRSX increased LC3-II and autophagic flux (reduced p62, polyQ80), while knockdown reduced LC3-II. Whether this secreted/autophagy function contributes to DHRSX-CDG pathophysiology, separate from the ER enzymatic role, is unresolved.

Cell types / anatomy involved: neurons (CL:0000540), cerebellar Purkinje cells (CL:0000121), and hepatocytes (CL:0000182) are plausibly affected by systemic hypoglycosylation, but cell-type-specific mechanistic data are not established for DHRSX-CDG.

7. Anatomical Structures Affected

Level	Structure	Ontology suggestion
Organ / system — nervous	Brain, cerebellum (atrophy), CNS, peripheral nerves	UBERON:0002037 (cerebellum), UBERON:0001017 (CNS)
Organ / system — musculoskeletal	Vertebral column (scoliosis), joints (contractures)	UBERON:0002391 (vertebral column)
Organ / system — sensory	Cochlea / inner ear (SNHL), eyes	UBERON:0001846 (inner ear), UBERON:0000970 (eye)
Organ / system — respiratory	Lungs / airway (variable insufficiency)	UBERON:0001004
Organ / system — gastrointestinal	GI tract (feeding difficulty, FTT)	UBERON:0005409
Craniofacial	Facial skeleton (dysmorphism)	—
Subcellular	ER membrane (LLO assembly site)	GO:0005789

Involvement is **bilateral** / **systemic**, consistent with a generalized biochemical defect rather than a focal lesion.

8. Temporal Development

- **Onset:** congenital / neonatal; insidious-to-chronic course.
- **Progression:** the encephalopathy is essentially static (developmental), while orthopedic features (scoliosis, joint contractures) are progressive during childhood.
- **Disease course:** chronic, lifelong.
- **Notable temporal biomarker feature:** the serum transferrin CDG-I abnormality can **normalize with age** — Patient 3's profile normalized by 17 months and was normal in his brother (Patient 4) — so a normal transferrin screen does not exclude the diagnosis ([PMID: 38821050](#)).
- **Critical periods / remission:** no spontaneous or treatment-induced clinical remission is described; the transferrin normalization is a biochemical, not clinical, phenomenon.

9. Inheritance and Population

- **Epidemiology:** Prevalence and incidence unknown; presumed ultra-rare (<1/1,000,000). Only 4 patients from 3 families reported worldwide (2024).
- **Inheritance pattern: pseudoautosomal-recessive** — the first described human disease of this class. Biallelic missense variants required; *DHRX* lies in PAR1 of X and Y and escapes X-inactivation ("The first and third steps are performed by *DHRX*, whose gene resides on the pseudoautosomal regions of the X and Y chromosomes" — [PMID: 38821050](#)).
- **Penetrance / expressivity:** presumed complete penetrance for biallelic LoF; variable expressivity (e.g., transferrin normalization, variable respiratory involvement).
- **Sex ratio:** ~1:1 — both sexes affected equally because PAR genes are biallelically expressed in both males and females. Reported cohort included two girls and two boys.
- **Consanguinity / founder effects:** homozygosity (consistent with consanguinity) present in the two singleton families; no founder allele established.
- **Carrier frequency / affected populations / geographic distribution:** not established.

10. Diagnostics

Biochemical screening. - Serum transferrin isoform analysis (IEF or HPLC/LC-MS): CDG **type I** pattern (defective ER-based N-glycan attachment) in patients 1–3. Caveat: can normalize with age or be normal (patient 4), so a normal screen does not rule out *DHRX*-CDG ([PMID: 38821050](#)). - **Polyisoprenoid profiling** (fibroblasts/tissue): elevated polyprenol with decreased dolichol and **no** polyprenal accumulation — the key biochemical discriminator from *SRD5A3*-CDG (which accumulates polyprenal / has a high polyprenol/dolichol ratio) ([PMID: 39395802](#); [PMID: 22304929](#)).

Genetic testing (definitive). Exome or genome sequencing of *DHRX*, with explicit attention to its **PAR1 (X/Y) location** — standard variant-calling pipelines may misannotate or under-cover pseudoautosomal loci. Single-gene/panel testing for CDG genes may include *DHRX* going forward.

Imaging: brain MRI may show cerebellar atrophy (HP:0001272).

Differential diagnosis: other dolichol-pathway/CDG-I disorders, especially **SRD5A3-CDG** (cerebello-ocular syndrome with eye malformations; distinguished by polyprenal accumulation) (PMID: 20852264); other CDG-I subtypes; and mitochondrial disorders, which clinically overlap with CDG (PMID: 29502919).

Screening: No newborn or population screening exists; diagnosis is currently by clinical suspicion → genetic testing. Note: transferrin-based dried-blood-spot screening exists for CDG generally (PMID: 30641270) but is limited here by the potentially normal transferrin profile.

11. Outcome / Prognosis

- **Severity/morbidity:** severe — profound intellectual disability, epilepsy, tube-dependent failure to thrive, sensorineural deafness, progressive scoliosis/contractures. High disease burden and dependency.
 - **Survival / mortality:** long-term survival and mortality data are **not available** given the 2024 discovery and tiny cohort.
 - **Prognostic factors / biomarkers:** none validated. Transferrin normalization is not known to predict clinical improvement.
 - **Recovery:** no recovery of neurodevelopmental deficits; supportive care only.
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12. Treatment

No disease-specific or targeted therapy exists. Management is supportive and multidisciplinary:

Problem	Supportive intervention	NCIT suggestion
Seizures	Anti-epileptic drugs	NCIT:C264 (Anticonvulsant Agent)
Failure to thrive	Nutritional support / gastrostomy tube feeding	NCIT:C173379 (Enteral Nutrition)
Hypotonia / scoliosis / contractures	Physiotherapy, orthopedic/spinal management, bracing	NCIT:C15515 (Physical Therapy)
Sensorineural hearing loss	Hearing aids / audiologic support	NCIT:C99290 (Hearing Aid)
Developmental disability	Early intervention, special education	—

Pharmacogenomics, gene/cell/RNA therapy, surgery-as-cure: none established. Unlike some CDG subtypes with dietary sugar therapy — **PGM1-CDG** (D-galactose; [PMID: 34043239](#)), **MPI-CDG** (mannose), or **SLC39A8-CDG** (galactose/manganese; [PMID: 34246313](#)) — dolichol-pathway CDGs including DHRSX-CDG have **no established sugar-supplement therapy**. Experimental/clinical-trial options are not yet available.

13. Prevention

- **Primary prevention:** not possible (monogenic congenital disorder).
- **Genetic counseling:** recommended for families. The pseudoautosomal-recessive mechanism has unusual implications — a pathogenic allele can be transmitted on either an X or a Y chromosome, and recurrence risk should be modeled accordingly (illustrated by the affected brothers who inherited one allele via the maternal X and one via the paternal Y).
- **Reproductive options:** prenatal diagnosis / preimplantation genetic testing are feasible in principle once familial variants are known, with careful handling of the PAR1 locus.
- **Secondary/tertiary prevention:** early multidisciplinary intervention to prevent complications (aspiration, contracture progression, scoliosis).

14. Other Species / Natural Disease

- **Taxonomy / orthologs:** The revised three-step dolichol pathway is **evolutionarily conserved**. In *Saccharomyces cerevisiae*, the dual role of human DHRSX is split between **two dedicated enzymes, Env9 and Tda5**; deletion of *ENV9* and *TDA5* causes accumulation of polyisoprenoid intermediates, transfer of immature LLOs onto nascent proteins, and defective N-glycosylation ([PMID: 42201967](#)).
- **Cellular model organism:** The classic **CHO Lec5 and Lec9** glycosylation-deficient mutants are natural DHRSX-null lines — the genomic region containing *DHRSX* is absent, and the defect is corrected by human *DHRSX* but not *SRD5A3* ([PMID: 39395802](#); [PMID: 38948797](#)).
- **Natural disease in companion animals / wildlife / zoonosis:** none described; not applicable.

15. Model Organisms

Model	Type	Nature	Recapitulation	Reference
CHO Lec5 / Lec9	Mammalian cell line	Natural DHRSX-null (genomic deletion)	Reproduces N-glycosylation defect, polyprenol ↑ / dolichol ↓ ; rescued by DHRSX not SRD5A3	PMID: 39395802 ; PMID: 38948797
<i>S. cerevisiae</i> <i>env9Δ tda5Δ</i>	Invertebrate/fungal	Double knockout of the two DHRSX orthologs	Recapitulates polyisoprenoid accumulation, immature LLO transfer, defective N-glycosylation	PMID: 42201967
DHRSX-KO cultured cells	In vitro	Engineered knockout	Reproduces metabolite signature (polyprenol ↑ , no polyprenal change)	PMID: 38821050

Complementation evidence: "N-glycan synthesis and changes in polyisoprenoid levels were corrected by complementation with human DHRSX but not with SRD5A3," and "Long-read whole genome sequencing of Lec5 and Lec9 cells did not reveal mutations in the ORF of SRD5A3, but the genomic region containing DHRSX was absent" ([PMID: 39395802](#)).

Limitations of models: cellular and yeast models capture the biochemical glycosylation defect but not the organismal neurodevelopmental phenotype. No dedicated mouse model of DHRSX-CDG is yet reported. Because rodents lack the human PAR1 arrangement of *DHRSX*, faithfully modeling the pseudoautosomal-recessive inheritance in mice is non-trivial.

Mechanistic Model / Interpretation

DHRSX-CDG unifies several previously disconnected observations into a single, coherent model that also rewrote a chapter of glycobiology. For decades, SRD5A3 was thought to be the sole "polyprenol reductase" converting polyprenol directly to dolichol. Two anomalies never fit: (1) SRD5A3-null patients retain substantial correctly glycosylated transferrin (~70%) and near-normal dolichol despite a supposed complete block ([PMID: 22304929](#)), implying an alternative route; and (2) the CHO Lec5/Lec9 mutants had a dolichol-formation defect with no SRD5A3 lesion.

The 2024 discovery resolves both: dolichol is made via a **three-step detour** in which DHRSX brackets SRD5A3. DHRSX first oxidizes polyprenol to polyprenal (activating the alkene), SRD5A3 reduces the activated alkene to dolichal, and DHRSX then reduces dolichal to dolichol. As stated succinctly in the CHO study: "dolichol synthesis from polyprenol occurs in three steps consisting of the conversion of polyprenol to polyprenal by DHRSX, the reduction of polyprenal to dolichal by SRD5A3, and the reduction of dolichal to dolichol, again by DHRSX" ([PMID: 39395802](#)). This reassigns SRD5A3 as a **polyprenal reductase** — a relabeling now echoed in SRD5A3-CDG proteomic studies ([PMID: 41732066](#); [PMID: 39360848](#)). The Lec5/Lec9 "mystery" is simply DHRSX-null CHO cells ([PMID: 38948797](#)), and the yeast Env9/Tda5 pair shows the detour is ancient and conserved ([PMID: 42201967](#)).

The **differential metabolite signatures** are the linchpin of both diagnosis and mechanism: DHRSX loss → polyprenol accumulates, polyprenal unchanged; SRD5A3 loss → polyprenal accumulates. This provides a clean biochemical discriminator between the two dolichol-pathway CDGs whose downstream glycosylation defects are otherwise indistinguishable.

Finally, the **genetics** are the most conceptually novel element. By residing in PAR1 on both X and Y and escaping X-inactivation, *DHRSX* behaves like an autosome but is physically sex-chromosomal — hence "pseudoautosomal-recessive." The affected brothers, carrying one variant from the maternal X and one from the paternal Y, are the definitive proof, and DHRSX-CDG is the first human disease established in this inheritance class.

Evidence Base

PMID	Title (abbrev.)	Role in this report
38821050	<i>A pseudoautosomal glycosylation disorder prompts the revision of dolichol biosynthesis</i>	Landmark. Defines DHRSX-CDG, the pseudoautosomal-recessive inheritance, the three-step detour, DHRSX dual specificity, and the 4-patient cohort
39395802	<i>Absence of DHRSX leads to N-glycosylation defects in Lec5/Lec9 CHO cells</i>	Establishes Lec5/Lec9 as DHRSX-null models; defines the polyprenol ↑ /dolichol ↓ /polyprenal-unchanged signature
38948797	<i>Lec5/Lec9 defect caused by absence of DHRSX</i>	Confirms genomic <i>DHRSX</i> absence in classic CHO mutants; restates the three enzymatic steps
42201967	<i>Revised three-step detour conserved in budding yeast</i>	Evolutionary conservation; Env9/Tda5 as yeast DHRSX orthologs
25076851	<i>DHRSX, a non-classical secretory protein associated with autophagy</i>	Documents DHRSX's second, enzyme-independent function in starvation-induced autophagy
22304929	<i>Life with too much polyprenol: polyprenol reductase deficiency</i>	SRD5A3-CDG comparator; residual glycosylation implies an alternative dolichol route (now explained by the detour)
20852264	<i>Cerebello-ocular syndrome due to dolichol metabolism (SRD5A3)</i>	Key differential diagnosis; dolichol-pathway CDG phenotype anchor
41732066	<i>Hypoglycosylation of serum N-glycoproteins in SRD5A3 deficiency</i>	Supports SRD5A3's reassignment as polyprenal reductase; downstream glycoproteomic consequences
39360848	<i>N-glycoproteomic/proteomic alterations in SRD5A3-deficient fibroblasts</i>	Illustrates systemic glycoprotein/organelle consequences of dolichol-pathway CDG
40902550	<i>Genetic disorders of dolichol synthesis and utilization (review)</i>	Contextual review situating DHRSX-CDG among dolichol CDGs
34043239	<i>D-galactose treatment monitoring in PGM1-CDG</i>	Contrast: treatable CDG subtype vs. untreatable dolichol CDGs
34246313	<i>SLC39A8-CDG clinical/ glycophenotype</i>	Contrast: treatable CDG (galactose/Mn); transferrin can normalize

PMID	Title (abbrev.)	Role in this report
29502919	<i>CDG vs mitochondrial disorder overlap</i>	Differential-diagnosis context
30641270	<i>Transferrin glycosylation from dried blood spots</i>	Screening methodology context
28139241	<i>Population-based CDG diagnosis (Spain)</i>	Epidemiologic/diagnostic context for CDG broadly
38811988	<i>RNA-seq in pediatric B-ALL (DHRSX duplication)</i>	Distinguishes an unrelated somatic DHRSX fusion from germline DHRSX-CDG

Limitations and Knowledge Gaps

1. **Tiny cohort.** All clinical data derive from 4 patients in 3 families ([PMID: 38821050](#)). Phenotype frequencies, penetrance, expressivity, natural history, and prognosis are therefore provisional. No expanded series exists (a caution: 9/10- or 6/7-type fractions in the literature belong to a separate PMM2-CDG cohort, not DHRSX-CDG).
2. **No epidemiology.** Prevalence, incidence, carrier frequency, founder effects, and geographic/ethnic distribution are unknown.
3. **Diagnostic caveat.** The transferrin CDG-I marker can normalize with age or be normal — meaning biochemical screening can miss cases; molecular testing at the PAR1 locus is essential.
4. **No survival/mortality data.**
5. **No therapy.** No disease-specific treatment, and no sugar-supplement therapy applicable (unlike PGM1-/MPI-/SLC39A8-CDG).
6. **Ontology gaps.** No dedicated MONDO, Orphanet, or ICD-11 entry yet; HPO frequency annotations for DHRSX-CDG are not curated.
7. **Unresolved dual function.** The contribution (if any) of DHRSX's secreted/autophagy-regulating role ([PMID: 25076851](#)) to disease pathophysiology, versus the ER dolichol-synthesis role, is unknown.
8. **No whole-organism animal model** faithfully reproducing the pseudoautosomal genetics and neurodevelopmental phenotype.

Proposed Follow-up Experiments / Actions

1. **International case-finding.** Reanalyze CDG-I cohorts and unsolved neurodevelopmental exomes/genomes with PAR1-aware variant calling to identify additional DHRSX-CDG patients and build a natural-history registry.
2. **Diagnostic standardization.** Deploy polyisoprenoid profiling (polyprenol/polyprenal/dolichol ratios) as a confirmatory second-tier test that discriminates DHRSX-CDG from SRD5A3-CDG independent of the unreliable transferrin marker.
3. **Ontology curation.** Request dedicated MONDO, Orphanet, and ICD-11 identifiers and HPO frequency annotations for DHRSX-CDG.
4. **Structure–function.** Determine DHRSX structure (cryo-EM/X-ray or refine AlphaFold model) and map the p.Val181Phe, p.Thr49Met, and p.Leu215Phe residues onto the NAD(P)(H)-binding and substrate pockets to establish genotype–function correlations.
5. **Animal/organoid models.** Generate humanized or conditional mouse and/or iPSC-derived neural organoid models to study tissue-specific vulnerability and to serve as therapeutic testbeds.
6. **Therapeutic exploration.** Test whether dolichol or dolichal supplementation, or bypass strategies, can restore LLO assembly in DHRSX-null cells — a first step toward a substrate-replacement approach.
7. **Resolve the dual-function question.** Use catalytically dead vs. secretion-deficient DHRSX constructs to dissect the dolichol-synthesis role from the autophagy/secretory role in disease-relevant cells.